
MOHAMED ALALWAN

AI Full-Stack Engineer | AI Engineer | Software Engineer

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PROFESSIONAL SUMMARY

AI Full-Stack Engineer with 3+ years of experience building production software, AI-powered applications, and enterprise systems. Experienced across Angular, NestJS, TypeScript, Python, MongoDB, and REST APIs, with hands-on work integrating LLM and agentic systems into SaaS products. Strong background in LangChain, LangSmith, RAG, document intelligence, API architecture, and cloud deployment across Google Cloud Platform (GCP) and Microsoft Azure.

PROFESSIONAL EXPERIENCE

AI Full-Stack Engineer

Penny Software | Seef, Bahrain | Apr 2026 - Present

- Build and ship AI-powered product features across Angular frontends, NestJS backends, MongoDB data models, and a Python-based AI service for Penny's procurement SaaS platform.
- Develop AI capabilities as an isolated Python SaaS service exposed through REST APIs, with backend API proxies connecting the service to Penny's main monorepo while keeping AI workloads decoupled from core application services.
- Build LLM and agentic workflows with LangChain, trace and monitor AI behavior with LangSmith, and integrate models from multiple providers, including self-hosted models running on Google Cloud Platform.
- Shipped AI Smart Scan for the procurement Bill module, extracting uploaded invoice data and helping users identify mismatches between invoice values and bill data already stored in the system.
- Built AI-assisted document extraction for RFQ and vendor offer workflows, allowing sellers to upload documents and automatically populate offer information from extracted values.
- Prototype customer-facing AI capabilities including Model Context Protocol (MCP), semantic search, image search, embeddings, and retrieval; contribute through peer code review, testing, QA, production releases, and task management in Plane.

AI Engineer | Senior Associate

KPMG Bahrain | Manama, Bahrain | Dec 2023 - Apr 2026

Promoted from Analyst to Associate to Senior Associate for technical delivery, client engagement, and contributions to the AI engineering team.

- Designed and delivered production AI systems using Azure OpenAI, Azure AI services, LangChain, Python, Retrieval-Augmented Generation (RAG), and agentic workflows across finance, banking, healthcare, tax, and government projects.
- Built AI solutions for document processing, payment validation, job matching, research automation, information extraction, and decision-support workflows.
- Evaluated OpenAI, Google Gemini, and Anthropic Claude models using prompt engineering, scenario-based testing, and accuracy evaluation to improve reliability and output quality.
- Delivered a production AI payment validation solution for a semi-government entity, achieving approximately 90% extraction accuracy and reducing manual monitoring workload by approximately 40%.
- Developed full-stack AI applications with Angular, React, Next.js, Node.js, Python/Flask, Azure Functions, Cosmos DB, and REST APIs; contributed to cloud architecture and served as technical lead on selected initiatives.

Software Engineer

AI Amthal Group | Sanabis, Bahrain | Apr 2023 - Nov 2023

- Developed responsive web and mobile applications using Angular, Ionic, TypeScript, JavaScript, HTML, CSS, and Bootstrap.
- Built backend services and REST APIs using Node.js, Express.js, ASP.NET, and C#, contributing to client-facing features, bug fixes, testing, and performance improvements.

IT Intern

Information & eGovernment Authority (iGA) | Muharraq, Bahrain | Jul 2022 - Jan 2023

- Supported enterprise IT systems and applications, assisting with software development, application support, testing, troubleshooting, and operational workflows in a government technology environment.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C#

Frontend: Angular, React, Next.js, Ionic, HTML, CSS, Bootstrap

Backend & APIs: NestJS, Node.js, Express.js, Flask, ASP.NET, REST APIs, API Integration, Serverless Functions

AI & LLM Engineering: Large Language Models (LLMs), Generative AI, Agentic AI, LangChain, LangSmith, Retrieval-Augmented Generation (RAG), Prompt Engineering, Document Intelligence, Model Evaluation, Embeddings, Vector Search, Semantic Search, Image Search, Model Context Protocol (MCP)

Models & Providers: OpenAI, Azure OpenAI, Google Gemini, Anthropic Claude, Self-Hosted Models

Databases: MongoDB, Azure Cosmos DB, PostgreSQL, SQL Server, MySQL

Cloud & DevOps: Google Cloud Platform (GCP), Microsoft Azure, Azure Functions, Azure AI Foundry, Azure DevOps, CI/CD, Git, GitHub

Engineering Practices: Full-Stack Development, SaaS Development, API Design, Software Architecture, Microservices, SDLC, Agile, Code Review, Quality Assurance, Testing, Debugging, Jira

EDUCATION

Bachelor of Information & Communication Technology: Programming

Bahrain Polytechnic | Isa Town, Bahrain | Feb 2023

Software Engineering Immersive

General Assembly | Manama, Bahrain | Apr 2023

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate | 2025

OutSystems Mobile Developer Specialist | 2024

OutSystems Associate Reactive Developer | 2023

General Assembly Software Engineering Immersive | 2023

OutSystems Associate Traditional Web Developer | 2023

Microsoft Certified: Azure Administrator Associate | 2021

VOLUNTEER EXPERIENCE

Volunteer Developer

Bahrain Diabetes Society | Bahrain | 2024 - 2025

- Contributed to an AI-powered diabetes care application supporting health awareness and patient engagement, and received recognition from the society for contributions to the initiative.

LANGUAGES

Arabic - Native | **English** - Professional Working Proficiency